Andrew Dillon

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Project Portfolio: andrewdillon.org

Education

California Polytechnic State University - San Luis Obispo (Cal Poly) **September 2024 – Present**
Bachelor of Science in Mechanical Engineering Expected Graduation: Spring 2028

- Dean's List: Winter 2025, Spring 2025, Fall 2025, Winter 2026, Spring 2026 Overall GPA: 3.96
- LOOP Scholarship (Class of 2028): Honors one engineering student per graduating class for academic and personal excellence

Coursework: Mechanical Systems Design, Thermodynamics, Fluid Mechanics, Mechatronics, System Dynamics, Manufacturing Processes, Mechanics of Materials, Materials Engineering, Engineering Data Analysis, Circuits, Engineering Programming.

Technical Skills

Mechanical Design: NX, SolidWorks, Fusion, GD&T, DFM/DFA, Functional Dimensioning, Tolerance Analysis, Mechanism Design
Analysis & Simulation: Ansys Mechanical, NX Nastran, MATLAB, Simulink, FEA, Fatigue Analysis, Mesh Convergence
Manufacturing & Test: CAM, CNC/Manual Machining, Metrology, Fixturing, Mechanical Assembly
Programming & Automation : Python, VBA, Motorola HC12X assembly language

Engineering Experience

Lam Research **June 2026 – August 2026**
Mechanical Engineering Intern *Fremont, CA*

- Owned end-to-end development of a precision lifting and alignment fixture from requirements definition and concept generation through down-selection, NX CAD modeling, analysis, and drawing release.
- Designed and released 7 new components, defining GD&T and limit dimensions from tolerance analyses to ensure fit, function, and manufacturability.
- Quantified 0.624-mm RSS misalignment at a critical interface using radial, horizontal, and clearance-driven rotational tolerance stacks; developed full-assembly contact FEA to validate structural integrity under maximum design load.
- Derived functional requirements and interface constraints from legacy hardware; coordinated changes to an adjacent assembly to ensure successful integration of new fixture with etch chamber.
- Propagated shared-component changes into 2 related fixtures to increase component commonality and reduce cost.

Cal Poly Wind Power **September 2024 – Present**
Overall Team Lead; previously Driveshaft Subsystem Lead and Outreach Lead *San Luis Obispo, CA*
Developing an autonomous scale wind turbine for the Collegiate Wind Competition; 1st Place, 2025 Western Region.

- Leading full-system integration across mechanical, electrical, and controls from design through manufacturing and testing.
- Designing and manufacturing a four-bar blade-pitch mechanism interfacing with an electromechanical linear actuator; modeling linkage backlash in MATLAB and evaluating turbine-level response to linkage-geometry change in Simulink.
- Assembling and testing the turbine in a wind tunnel, varying blade pitch and applied load while collecting RPM and power data to characterize turbine performance and establish operating setpoints for autonomous power optimization and curtailment.
- Designed, analyzed, and manufactured a precision driveshaft subsystem using MATLAB shaft-stress and bearing-friction models, Ansys FEA, and a custom machining fixture; achieved 0.0015-in total runout relative to the main hub.
- Led outreach and professional networking events; organized member support for a $3M offshore wind funding package.

Cal Poly Machine Shops **September 2025 – Present**
Senior Shop Technician *San Luis Obispo, CA*

- Training students on lathes, mills, welding, laser, 3D printing; giving technical advice on design for manufacturability.
- Performing diagnostics and maintenance on Bridgeport and Clausing machines, addressing belt and bearing issues.

Nova Group, Inc. **June 2025 – September 2025**
Operations Intern *Napa, CA | Tokyo, JP*

- Developed Python/VBA tools to automate image geo-tagging in Google Earth and specification reference linking in Excel.
- Red-lined P&ID sheets per MIL-STD and performed pipe/fitting takeoffs to identify flaws in estimation process.
- Supported daily operations across three defense infrastructure projects ($225M total value), tracking status and invoices.

Cal Poly Industrial and Manufacturing Engineering **January 2026 – June 2026**
Teaching Assistant *San Luis Obispo, CA*

- Instructed students in machine tool operation, metrology practices, and technical drawing interpretation during hands-on labs.

Projects

Custom Vehicle Storage Drawer System | *DFM, Packaging Design, Load-Bearing Hardware Selection*

- Designed custom-fit drawer system from measured trunk geometry and clearance constraints, selecting load-rated hardware.

Heat-Treated Carabiner | *DFM, Spring Mechanism Design, CAD/CAM, CNC Mill*

- Designed a functional spring-loaded carabiner by developing gate geometry and tuning spring behavior for reliable operation.

Heat-Treated Bolt-Action Pens | *DFM, Kinematic Constraints, CAD/CAM, Manual Lathe, CNC Mill*

- Developed a compact bolt-action mechanism built around precise internal travel, ink cartridge fit, and a seamless exterior.

Anodized CNC Chess Sets | *DFM, Repeatable Production, Anodizing, CAD/CAM, CNC Lathe, CNC Mill*

- Produced two 32-piece chess sets using repeatable machining setups and processes for consistent geometry and finish.